AI-Assisted ADR Reporting & Institutional Verification System

Claude Code Master Build Prompt | Full Product Specification | PvPI Form Version 1.4

1. Project overview

Build a complete responsive web application for AI-assisted adverse drug reaction (ADR) reporting, clinical documentation, institutional verification, and submission management. Intended primarily for hospitals, clinical pharmacy and pharmacy practice departments, and institutional pharmacovigilance units. Use ADRRF_2.pdf (Indian Pharmacopoeia Commission Suspected ADR Reporting Form, Version 1.4) as the reference for report fields and layout.

Core principle: AI assists extraction, questioning, summarization and consistency checking. It must never invent clinical information, independently confirm causality, or replace authorized professional verification. Build a serious hospital information system, not a generic chatbot.

2. Technology stack

- Frontend: Next.js App Router, TypeScript, Tailwind CSS, shadcn/ui, Lucide, React Hook Form and Zod.
- Backend: Next.js server routes/actions, PostgreSQL, Prisma ORM, secure authentication, server-side PDF generation and private document storage.
- AI: provider-independent multimodal extraction service with structured JSON outputs validated by Zod; separate extraction, adaptive-question, narrative and consistency services.
- Use environment variables for secrets, server-side authorization, secure APIs, and synthetic or appropriately de-identified demonstration data.

3. UI and visual design

Simple, restrained, clinical SaaS design. Use Inter (system sans-serif fallback), predominantly 400/500/600 weights. Page titles 24-28px, section headings 16-18px, body 14px, labels 13-14px, table text 13-14px. White #FFFFFF, secondary #F8FAFC, text #0F172A, muted #64748B, border #E2E8F0, accent blue #2563EB, success #15803D, warning #B45309, error #B91C1C.

No gradients, glassmorphism, neon, unnecessary illustrations, oversized radii, heavy shadows, cluttered analytics, decorative animations or emojis. Use 6-10px radii, subtle dividers and consistent spacing. Desktop left sidebar: Dashboard, New ADR Report, My Reports, Review Queue (authorized users), Submitted Reports, Notifications, Settings. Compact top bar: institution, workspace, notifications, profile, logout. Mobile collapsible drawer.

4. Authentication, identities and roles

Reporting healthcare professional

- Clinician, pharmacist, nurse, physician or other institutionally authorized healthcare professional.
- Create reports, upload evidence, choose either reporting mode, save/edit drafts, submit for review, track status, respond to comments and resubmit corrections.

Authorized ADR reviewer

- HoD of Pharmacy Practice, Chief Pharmacist, AMC coordinator or other designated official; authorization is explicitly assigned, not automatically inferred from title.

- View shared institutional queue, claim cases, inspect original sources, verify data, comment, request correction, hold, approve for forwarding and view audit history.
- Support 6-7 or more reviewers without an artificial cap.

Reporting coordinator and institutional administrator

- Coordinator: manage approved reports, authorized external forwarding, acknowledgements and submission tracking.
- Administrator: manage accounts, roles, departments, notification rules, institutional settings and permitted audit access.

Patient or caregiver participation

Allow guided patient-provided information. For the hospital-focused MVP, treat such input as a draft/information submission requiring an appropriate professional review or patient-reporting pathway. Never attach a professional signature automatically to a patient-generated report. Keep reporting identity separate from interface choice.

5. Dashboard

- Small metric cards: total, draft, pending, returned, approved, externally submitted, potentially serious.
- Recent cases table: report ID, patient reference, suspected drug/reaction, reporter, submission date, status, action. Search, filters and sorting.
- Quick actions: New ADR Report, Continue Draft, My Reports and Review Queue (if authorized).

6. Two reporting experiences: Structured and Guided

Users choose Structured Reporting or Guided Reporting based on workflow preference, not perceived expertise. Both modes populate the same normalized clinical schema. Users can switch without losing entered information. Allow contextual guided help from structured fields and advanced details within guided mode.

Structured Reporting - form sections

- A. Patient: initials, age at reaction onset or DOB, gender, weight and optional institutional reference.
- B. Reaction: start/stop dates, description, management, dated investigations, medical/medication history, allergies, relevant pregnancy, hepatic/renal information, seriousness and outcome.
- C. Suspected medicines (repeatable): brand/generic, manufacturer, batch/lot, expiry, dose, route, frequency, start/stop dates, indication, actions after reaction, rechallenge details, and authorized causality assessment when supplied.
- Concomitant products (repeatable): medicines including OTC/herbal/self-medication, dose, route, frequency, dates and indication; distinguish products used to treat the reaction.
- D. Reporter: verified account name, address, PIN, email, phone and occupation; support profile correction. Login is not automatically a legally accepted signature.

Guided Reporting - adaptive questions

- Step-by-step questions in groups of roughly 2-4 when appropriate, with multiple choice, dates, yes/no, free text, optional explanations and uploads.
- Use plain-language equivalents of dechallenge, hospitalization and reaction chronology; e.g. "After the medicine was stopped, did the symptoms improve?"
- Include "I do not know", approximate dates and review/correction options; never force fabricated answers or encourage medicine rechallenge.
- Use a validated question bank mapped to PvPI fields; AI selects follow-ups for missing mandatory data, unclear history, contradictory dates and possible seriousness. Do not let AI invent clinical criteria.

7. Upload and document extraction

- Optional JPG, PNG and PDF uploads: prescriptions, packaging, strips, labs, discharge summaries, clinical notes and supporting documents. Drag/drop, preview, progress, remove, secure storage and validation.
- Extract drug names, strength, dose, route, frequency, manufacturer, batch/expiry, patient details, dated investigations, findings and dates.
- Each extracted value retains document source, evidence reference and unverified/verified status; require confirmation for uncertain handwriting.

8. Shared clinical data engine

Normalize each field as value, type, provenance, source reference, optional extraction confidence, verification state, modified-by and timestamp, and unknown/missing status. Distinguish user-provided, document-extracted, clinician-verified, AI-suggested and unknown. Never overwrite verified values silently. Preserve original responses and corrections.

9. AI narrative generation

Convert fragmented clinical notes into concise, professional narratives without adding unsupported diagnoses, dates, lab results or causal conclusions. Example: "Started amoxicillin 500 mg TID on 12 October; rash began 14 October; stopped 15 October; cetirizine given; improved by 17 October" becomes a factual chronological narrative preserving these details. Allow manual editing and show material differences between inputs and generated text.

10. Completeness and consistency checks

- Check mandatory patient, onset, reaction, suspected medicine and reporter information.
- Flag onset before drug start, contradictory dates/doses, duplicates, unclear outcomes, incomplete seriousness, conflicts with uploads and absent relevant history.
- Offer correct, confirm as recorded, explain or mark uncertain; never silently change a clinical fact.
- Display mandatory completeness, optional gaps, unverified items and unresolved contradictions; completion percentage is not clinical validity.

11. PvPI-aligned PDF preview and generation

- Preserve the reference form sections A-D and provide draft preview, download, print, finalized version and extra pages as necessary.
- Mark unknown data as unknown/unreported. Never invent AMC numbers, worldwide unique IDs, receiving-personnel signatures or causality assessments.
- Include internal case/version reference outside official fields where appropriate. Provide reporter attestation and printable handwritten-signature option.
- Treat electronic signing as configurable only after institutional/receiving-authority acceptance is established.

12. Institutional review queue

After "Submit for Institutional Review", case enters a centralized queue visible to all authorized reviewers. Table: ID, patient reference, suspected medicine, reaction, reporter, date, seriousness flag, assignee, status and action. Filters: pending, unassigned, assigned to me, potentially serious, returned, approved, on hold.

"Take Up This Case" claims the case, records reviewer/time, prevents conflicting decisions using transactional assignment and permits authorized reassignment. Other permitted reviewers retain visibility. One authorized reviewer may approve routine reports according to institutional policy; do not require unanimous approval.

13. Reviewer detail screen

- Header: case ID, status, suspected reaction, reporter, submitted date, assignee and seriousness indicator.
- Desktop two-column workspace: compiled ADR information on left; originals, questionnaire responses, extraction provenance and source references on right. Click a field to inspect its supporting evidence.
- Checklist: patient data; reaction chronology; suspected medication; clinical evidence; seriousness/outcome; reporter/mandatory completeness.
- Timestamped, attributed general and field-specific comments; request information, record discrepancies and reasons for return.
- Controls: Approve for Forwarding, Return for Correction, Place on Hold, Request Second Opinion. Record decision and actor; approval is not proof of drug causality.
- Require configured checks and acknowledgment/resolution of significant discrepancies. Highlight changes in resubmitted versions.

14. Shared reviewers and availability

Support a shared queue for 6-7+ explicitly authorized officials, available through the application 24/7. Notify new cases, priority cases, assignment and correction. Allow reassignments, second opinions and escalation. 24/7 application availability does not imply continuous human staffing; configure an on-call/escalation rota where needed.

15. Potentially serious cases

Flag possible death, life-threatening reaction, initial/prolonged hospitalization, disability, congenital anomaly or other medically important event. Treat AI flags as preliminary. Immediately alert designated personnel, prioritize queue position, timestamp alerts, support alternate reviewers and track time since reporting. Internal approval must not obstruct applicable urgent reporting duties. Timelines and escalation rules are institution-configurable, not invented universal deadlines.

16. Report status machine

- Draft; Ready for Review; Submitted for Institutional Review; Under Review; Returned for Correction; Resubmitted; On Hold; Institutionally Approved; Ready for External Submission; Externally Submitted; Acknowledged (if applicable).
- Track external method, destination, submitting official, date, external reference and acknowledgment where available. Printing or downloading does not equal submission. Never simulate real regulatory submission.

17. Notifications

Reporter alerts: returned, information requested, approved, submission status changed. Reviewer alerts: new case, serious flag, reassignment, resubmission. Do not expose identifiable patient information in notification previews without authorization.

18. Audit trail and versioning

Record creation, uploads, extraction, AI narratives, human edits, reviewer assignment/comments/decisions, return, approval, external forwarding and acknowledgment. Store actor, timestamp, case ID and relevant before/after values. Preserve immutable finalized versions and support version comparison. Ordinary users cannot alter audit history.

19. Security and privacy

Strong authentication, secure sessions, institution-level data isolation, least privilege, API/database authorization, encrypted transport/storage, private documents, upload scanning/validation, rate limiting, secure resets, expiration and audit logs. Do not send identifiable patient data to external AI providers without approved arrangements. Use synthetic/de-identified cases for MVP. Do not claim legal compliance without institutional/legal assessment.

20. Database models

Institutions, departments, users, roles, assignments, patient references, ADR cases, suspected and concomitant medicines, reactions, medical histories, investigations, uploaded documents, extracted fields, answers, AI narratives, verification checks, reviewer assignments/comments/decisions, report versions, external submissions, notifications and audit logs. Add relationships, constraints and indexes; enforce institution isolation.

21. Application pages

Login; Dashboard; New Report; Reporting Mode Selection; Structured Reporting; Guided Reporting; Upload/Extraction; Report Preview; My Reports; Review Queue; Reviewer Detail; Approved Reports; External Submission; Notifications; Profile; Role Administration; Audit History. All pages responsive with loading, empty/error states, validation, save feedback, unsaved-change warnings and keyboard accessibility.

22. Demonstration dataset

Fictional institution and accounts: pharmacist, physician, reviewer, coordinator, administrator. At least five fictional ADR cases spanning draft, pending, returned, approved and potentially serious. ADR-001 is a fictional suspected cutaneous reaction. Clearly mark all demonstration data.

23. Build strategy and implementation priorities

IMPORTANT: Build the workflow before the AI features

For the first working version, PRIORITIZE these four components: (1) healthcare professional reporting interface, (2) shared institutional reviewer queue, (3) reviewer verification/detail screen, and (4) PvPI-aligned PDF generation. Together they demonstrate the central institutional ADR reporting workflow without dependence on AI services.

Only AFTER these four components are working end-to-end should you add AI document extraction and the Guided Reporting questionnaire. This order lets us verify the actual reporting, review, correction and forwarding workflow before introducing additional AI complexity. Do not build attractive but disconnected screens.

Phase 1 - Functional workflow MVP

- Project setup, professional design system, authentication, roles, database and dashboard.
- Healthcare professional structured form, validation, save/resume drafts and institutional submission.
- Shared review queue, claim/reassign, reviewer detail screen, checklist, comments, return/approve and resubmission.
- PvPI PDF preview/download/print, attestation, audit trail and distinct external-submission tracking.
- Verify end-to-end with fictional cases and permission tests before proceeding.

Phase 2 - AI-assisted reporting

- Add optional image/document upload and structured extraction with provenance.
- Add Guided Reporting with validated adaptive question bank and mode switching.
- Add AI narrative compilation, consistency checking, missing-field prompts and user confirmation.

Phase 3 - Institutional operations

- Notifications, escalation for potentially serious reports, second opinions, assignment handoffs, enhanced audit/version comparison and coordinator tools.

Phase 4 - Validation and refinement

- Unit/integration/security tests, role-access tests, PDF field mapping tests, clinical scenario testing, accessibility and mobile checks.
- Assess time to complete, mandatory-field completeness, clinically significant errors, correction burden and usability.

24. Acceptance criteria

- Authenticated professional creates/saves/submits a report and receives review decisions.
- Structured and Guided modes eventually share one case schema; mode switching preserves answers.
- Uploaded documents produce reviewable source-linked extracted fields; missing mandatory information is flagged.
- AI narratives do not invent facts; users can review/edit outputs.
- PvPI-aligned draft PDF can be generated, and institutional approval is distinct from external submission.
- Multiple authorized reviewers can view queue, claim cases safely, inspect evidence, comment, approve/return/hold and request second opinion.
- Returned reports can be corrected and resubmitted with preserved versions.
- Seriousness flags and escalation work; permissions, privacy boundaries and audit logs are enforced.
- The UI is minimal, responsive, accessible and genuinely usable, not merely a static mockup.

25. Final instructions to Claude Code

Inspect the project directory and ADRRF_2.pdf first. Produce a concise architecture and implementation plan. Then build the functional workflow MVP in Phase 1 and test it end-to-end before adding AI features. Implement real persistence, navigation, validation, permissions and PDF generation. If an AI provider is not configured, display an honest unavailable state and permit manual entry. Never fabricate external submissions or clinical facts. Keep modules maintainable, test each phase, and resolve errors before proceeding.

Design principle: Make ADR reporting easier without making clinical documentation less trustworthy.